

NEXA NANO-FABRICATION FACILITY

EQUIPMENT CALIBRATION & VERIFICATION SHEET

Bay 3 – Thin Film Photolithography Zone | Metrology Core B

✓ CALIBRATION PASSED

Status: APPROVED FOR USE

Valid: 24-Jul-2026 to 24-Jan-2027

SECTION 1: EQUIPMENT IDENTIFICATION & SPECIFICATIONS

Equipment Name:	OptiSpec 5000 Laser Ellipsometer	Asset Control ID:	NFX-EQ-ELL-094
Manufacturer:	Meridian Precision Instruments Inc.	Serial Number:	MPI-2023-8849-X
Model Number:	OS-5000-DUV-HighRes	Cleanroom Location:	Bay 3 / Workstation B-02
Measurement Range:	0.5 nm – 5,000.0 nm	Cleanroom Class:	ISO 5 (Class 100 Standard)
Laser Wavelengths:	632.8 nm (HeNe), 355.0 nm (UV)	Calibration Cycle:	6 Months (Semi-Annual)
Primary Contact:	Dr. Aris Thorne (Metrology Lead)	Department:	Microfabrication Systems

SECTION 2: ENVIRONMENTAL MONITORING DURING CALIBRATION

Ambient Temperature:	20.1 °C (±0.3 °C limit)	Relative Humidity:	42.5% RH (±3.0% limit)
Laminar Airflow:	0.45 m/s (HEPA Filtration)	Particle Count (0.5 µm):	280 particles/m³ (Pass ISO 5)
Vibration Control:	Active Isolation Table ON	Barometric Pressure:	1013.2 mbar (Nominal)

SECTION 3: TRACEABLE CALIBRATION REFERENCE STANDARDS

Standard ID	Description	Nominal Value / Spec	NIST / ISO Cert #	Cal Date	DueDate
NFX-STD-SiO2-10	Thermal SiO ₂ Standard Wafer	14.8 nm ± 0.15 nm	NIST-98421-B	10-Jan-2026	10-Jan-2027
NFX-STD-SiO2-100	Thermal SiO ₂ Standard Wafer	101.4 nm ± 0.20 nm	NIST-98422-B	10-Jan-2026	10-Jan-2027
NFX-STD-SiO2-1000	Thermal SiO ₂ Standard Wafer	1002.5 nm ± 0.50 nm	NIST-98425-C	15-Feb-2026	15-Feb-2027
NFX-STD-Si-Rl	Single Crystal Si Standard	$n = 3.8650 @ 632.8 \text{ nm}$	ISO-17025-882	04-Mar-2026	04-Mar-2027

SECTION 4: MULTI-POINT MEASUREMENT RESULTS (AS-FOUND VS. AS-LEFT)

Parameter Test	Reference	As-Found	As-Left	Tolerance	Status
Ultra-Thin SiO ₂ Film Thickness	14.80 nm	15.02 nm	14.83 nm	± 0.15 nm	✓ PASS
Medium SiO ₂ Oxide Film Thickness	101.40 nm	101.95 nm	101.44 nm	± 0.30 nm	✓ PASS
Thick SiO ₂ Oxide Film Thickness	1002.50 nm	1004.10 nm	1002.72 nm	± 1.00 nm	✓ PASS
Refractive Index (n) @ 632.8nm	1.4570	1.4592	1.4572	± 0.0010	✓ PASS
Incident Angle Calibration (θ)	70.000°	70.045°	70.002°	± 0.010°	✓ PASS
Polarizer Zero Angle Alignment	0.000°	0.018°	0.001°	± 0.005°	✓ PASS
Laser Power Stability (632.8nm Output)	5.00 mW	4.88 mW	4.98 mW	± 0.10 mW	✓ PASS
Sample Stage Repeatability (X/Y Axis)	0.000 µm	0.420 µm	0.080 µm	± 0.250 µm	✓ PASS